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(1) Pulse Input = 1μS

i) 10 MHz

$t_r = \frac{0.35}{f_{3dB}} = \frac{0.35}{10\times10^6}$

$= 35\,nS$

$V_{out} = 1(1-e^{-T/\tau})$

$= 1-e^{-1\mu S/15.9nS}$

$= 1-e^{-62.9}$

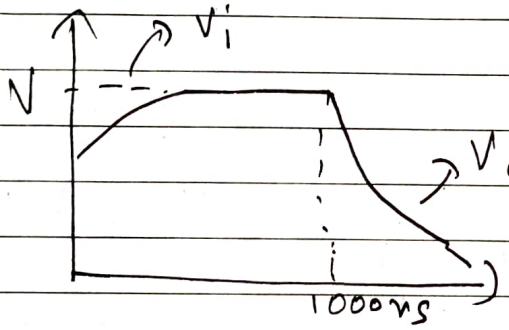
≈ 1

Input pulse width(T): 1μS = 1000 nS

Rise Time(t_r) = 35 nS

$t_r << T$

$V_{out} = \begin{cases} 1-e^{t/\tau} & 0 \le t \le 1\mu S \\ e^{-(t-1)/\tau} & t \ge 1\mu S \end{cases}$



ii) 0.1 MHz

rise time, $t_r = \frac{0.35}{f_{3dB}} = \frac{0.35}{0.1\times10^6} = 3.5\times10^{-6}\,s$

$= 3.5\mu S$

Input pulse width(T) = 1 μs

Rise time(t_r) = 3.5 μs

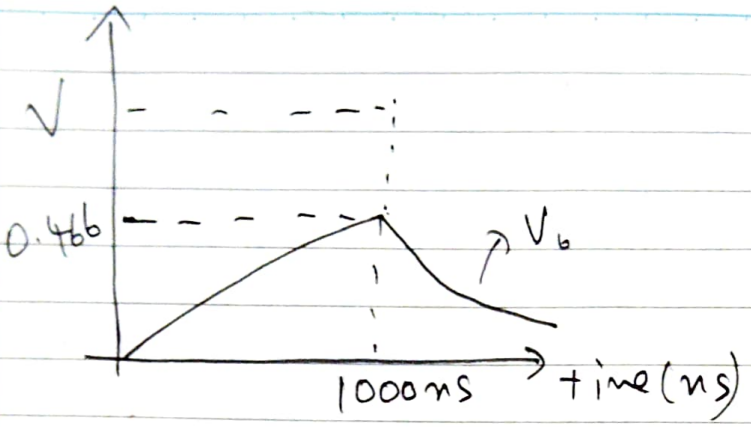
$t_r >> T$

$V_{out} = 1(1-e^{-T/\tau})$

$= 1-e^{-0.6289}$

$= 1-0.5334$

$\approx 0.4666\,V$



(2)

$f = 1KHz$

$\Rightarrow T = 1ms$

$V_{pR} = 10\ V$

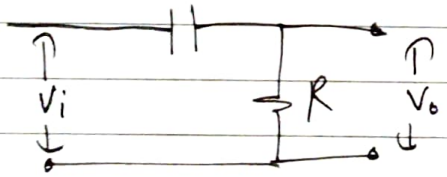
$\tau = 1ms = RC$

$V_{out}(t) = V_R(t)$

$= V_{in}(t) - V_c(t)$

$V_{in} = V_c + V_R$

(i) High pass ckt



(a) $t = T/2$

$V_c(T/2)$ will be V_2

$V_2 = 10 + (V_1 - 10)e^{-T/2\tau}$

due to Symetry $V_2 = -V_1$

$-V_1 = 10 + (V_1 - 10)e^{-0.5}$

$V_1 = \frac{10(e^{-0.5} - 1)}{e^{-0.5} + 1} \approx -2.45$

So capacitor V swings b/w $V_1 = -2.45\ V$ & $V_2 = +2.45\ V$

$V_{out} = V_R = V_{in} - V_c$

@ $t=0^+$, $V_{in}=10V$

$V_c = V_1 = -2.45\ V$ (cannot change instantaneously)

$V_{out} = 10-(-2.45) = +12.45$

@ $t=T^-$

$V_{in} = -10$

$V_c = V_1 = -2.45$

$V_{out} = -10-(-2.45)$

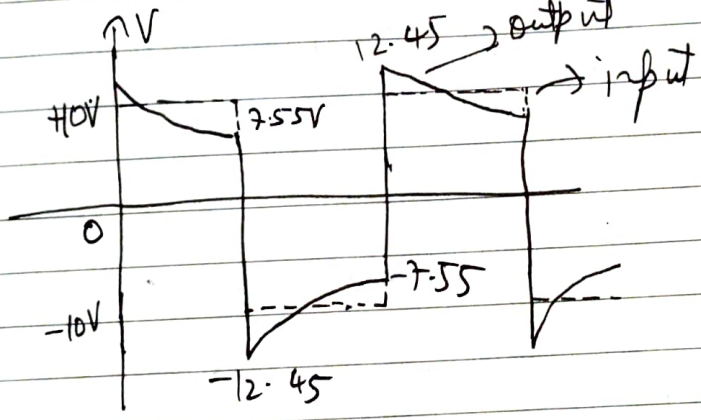
$= -7.55$

@ $t=(T/2)^-$, $V_{in}=10V$

V_c chages to $V_2 = +2.45$

$V_{out} = 10-2.45 = 7.55$ (decays)

$V_{out} = 12.45\ e^{-0.5} \approx 7.55\ V$



ii) Low pass filter

$T = 1/f = 10^{-3}\ s$ $RC = C = 10^{-3}\ s$

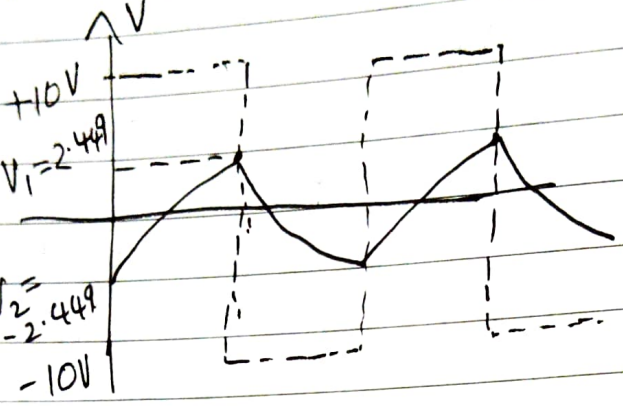
$V = 20$

$$V_1 = \frac{V}{2}\left(\frac{1-e^{-T/2RC}}{1+e^{-T/2RC}}\right) = \frac{20}{2}\left(\frac{1-e^{-0.5}}{1+e^{+0.5}}\right)$$

$= 2.449\ V$

$V_2 = -V_1 = -2.449\ V$

Peak to peak values of o/p = 4.898V



(3) f = 10kHz

tilt = 3.8% R = ?

C = 0.47 uF

$T = \frac{1}{f} = 100\mu s$

$t_p = T/2 = 50\mu s$

$\% tilt = \frac{t_p}{\tau} \times 100$

$3.8 = \frac{50\times10^{-6}}{\tau} \times 100$

$\tau = 1.316 ms$

$f_L = \frac{1}{2\pi\tau} = \frac{1}{2\pi \times 1.316\times10^{-3}} \approx 121Hz$

$\tau = R \times C$

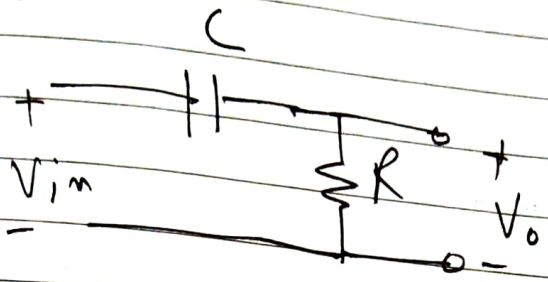
$R = \tau / C$

~~$= 1.316\times10^{-3}$~~

$R = \frac{1.316\times10^{-3}}{0.47\times10^{-6}}$

$R = 2799.6\Omega \approx 2.8k\Omega$

(4) Ex)



$$V_o = \frac{Z_R}{Z_R + Z_C} V_{in}$$

$$\frac{V_o}{V_{in}} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC}$$

If $\omega RC << 1$ (low frequencies)

$$\left[\omega << \frac{1}{RC}\right]$$

$$V_o = j\omega RC V_{in}$$

$$V_o = RC \frac{dV_{in}}{dt}$$ which is the output of a differentiator

For a good differentiator

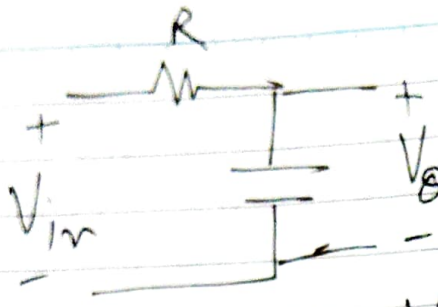
$$\boxed{RC << 1} \text{ or } \boxed{\omega << \frac{1}{RC}}$$

$\Rightarrow T <<$ time pd. of the signal

If $T << T_{input}$

$$V_{out} = RC \frac{dV_i}{dt}$$

(5)



$$\frac{V_{out}}{V_{in}} = \frac{1/j\omega C}{1/j\omega C + R} = \frac{1}{1 + j\omega RC}$$

Considering $|j\omega RC| >> 1$ ie $\boxed{\omega >> \frac{1}{RC}}$

we get $V_{out} = V_{in} \times \frac{1}{j\omega RC}$

$$\Rightarrow \boxed{V_{out}(t) = \frac{1}{RC} \int V_{in}(t)\, dt}$$

So for good integrator $\omega RC >> 1$

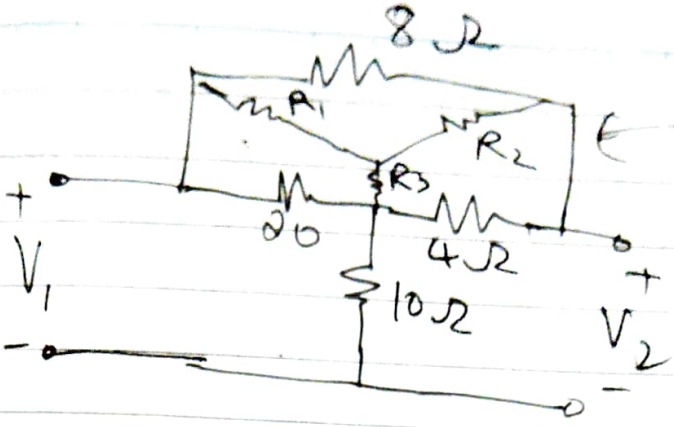
$$RC >> \frac{1}{\omega}$$

$$\Rightarrow RC >> T_{signal}$$

∴ Time constant very greater than time pd. of input

∴ if $\tau >> T_{input}$

$$\boxed{V_{out} = \frac{1}{RC} \int V_{in}\, dt}$$

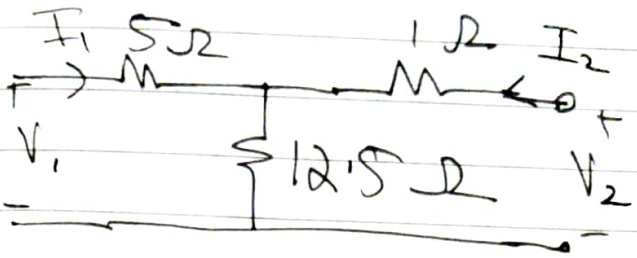


Delta to Star

$R_1 = \frac{160}{32} = 5\ \Omega$

$R_2 = \frac{32}{32} = 1\ \Omega$

$R_3 = \frac{80}{32} = 2.5\ \Omega$



Applying KVL

$$V_1 - 5I_1 - 12.5(I_1 + I_2) = 0$$

$$V_1 = 17.5I_1 + 12.5I_2$$

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = 12.5I_1 + 13.5I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

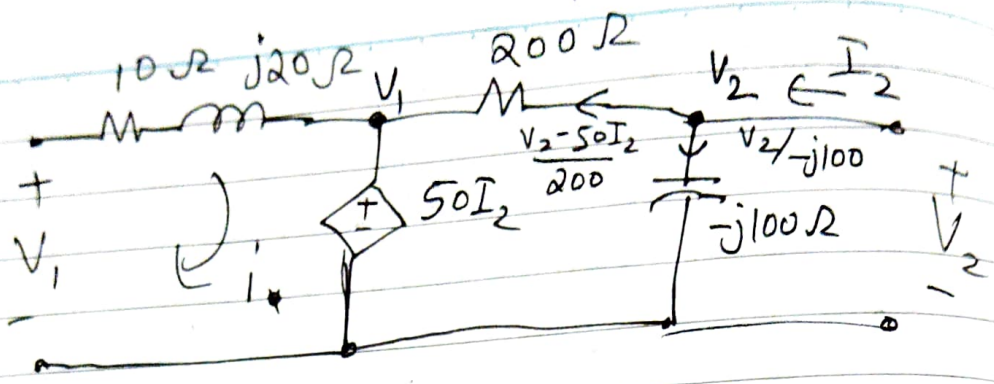
$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 17.5 & 12.5 \\ 12.5 & 13.5 \end{bmatrix}$$

$$[Y] = \begin{bmatrix} +Z_{22}/\Delta Z & -Z_{12}/\Delta Z \\ -Z_{21}/\Delta Z & +Z_{11}/\Delta Z \end{bmatrix}$$

$$= \begin{bmatrix} 13.5/80 & -12.5/80 \\ -12.5/80 & 17.5/80 \end{bmatrix} = \begin{bmatrix} 0.16875 & -0.15625 \\ -0.15625 & 0.21875 \end{bmatrix}$$

(7)

$V_1 = 50 I_2$



KCL at node 2

$$I_2 = \frac{V_2 - 50I_2}{200} - \frac{V_2}{j100}$$

$$I_2\left[1 + \frac{1}{4}\right] = V_2\left[\frac{1}{200} - \frac{1}{j100}\right]$$

$$1.25 I_2 = V_2\left(\frac{1+2j}{200}\right)$$

$$V_2 = \frac{250 I_2}{1+2j} \times \frac{1-2j}{1-2j}$$

$$V_2 = 50 I_2 (1-2j)$$

$$V_2 = (50 - 100j) I_2 \quad -(2)$$

KVL at loop 1

$$V_1 - (10+20j) I_1 - 50 I_2 = 0$$

$$V_1 = (10+20j) I_1 + 50 I_2 \quad -(1)$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad -(3)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad -(4)$$

Comparing (1), (2), (3), (4)

$Z_{11} = (10+20j)\Omega \quad Z_{12} = 50\Omega$

$Z_{21} = 0 \quad Z_{22} = (50-100j)\Omega$

$$[h] = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$\Delta Z = (10+20j)(50-100j)$$

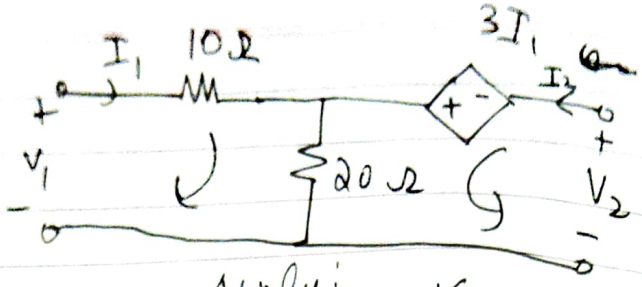
$$= 500 + 2000 + 1000j - 1000j$$

$$\Delta Z = 2500$$

$$[h] = \begin{bmatrix} \frac{2500 \times (50+100j)}{(50-100j)(50+100j)} & \frac{50}{50-100j} \times \frac{50+100j}{50+100j} \\ 0 & \frac{1}{50-100j} \end{bmatrix}$$

$$= \begin{bmatrix} (10+20j)\Omega & 0.2+0.4j \\ 0 & (0.004+0.008j) \\ & S \end{bmatrix}$$

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Applying KVL

$V_1 - 10I_1 - 20(I_1 + I_2) = 0$

$V_1 = 30I_1 + 20I_2$ ①

$V_2 + 3I_1 - 20(I_1 + I_2) = 0$

$V_2 = 17I_1 + 20I_2$ ②

$V_1 = Z_{11}I_1 + Z_{12}I_2$ ③

$V_2 = Z_{21}I_1 + Z_{22}I_2$ ④

Comparing ①, ②, ③, ④

$Z_{11} = 30 \quad Z_{12} = 20$

$Z_{21} = 17 \quad Z_{22} = 20$

$\Delta Z = 260$

$$[T] = \begin{bmatrix} Z_{11}/Z_{21} & \Delta Z/Z_{21} \\ 1/Z_{21} & Z_{22}/Z_{11} \end{bmatrix}$$

$$[T] = \begin{bmatrix} 30/17 & 260/17\ \Omega \\ 1/17\ S & 20/17 \end{bmatrix}$$

$$[T] = \begin{bmatrix} 10 & 1.5\,\Omega \\ 2S & 4 \end{bmatrix}$$

$$[z] = \begin{bmatrix} A/C & \Delta T/C \\ 1/C & D/C \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 18.5 \\ 0.5 & 2 \end{bmatrix} \Omega$$

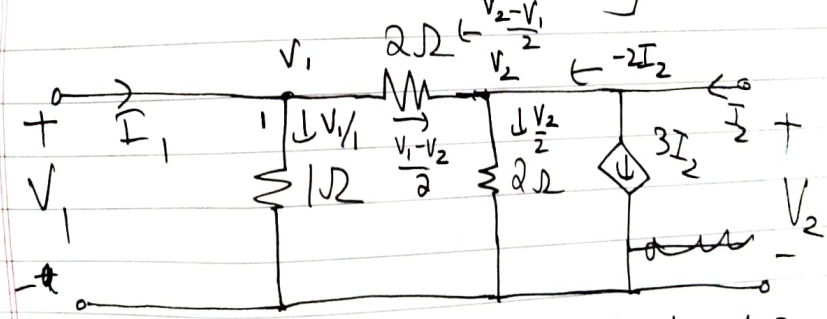
$\Delta T = 40 - 3 = 37$

$A = 10$
$B = 1.5$
$C = 2$
$D = 4$

$$[g] = \begin{bmatrix} C/A & -\Delta T/A \\ 1/A & B/A \end{bmatrix}$$

$$= \begin{bmatrix} 0.2\,S & -3.7 \\ 0.1 & 0.15\,\Omega \end{bmatrix}$$

(10)



KCL at Node 1

$$I_1 = \frac{V_1 - V_2}{2} + V_1$$

$$I_1 = 1.5V_1 - 0.5V_2 \quad (1)$$

KCL at Node 2

$$-2I_2 = \frac{V_2 - V_1}{2} + \frac{V_2}{2}$$

$$I_2 = 0.25V_1 - 0.5V_2 \quad (2)$$

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad (3)$$

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Comparing ①, ②, ③, ④

$$[Y] = \begin{bmatrix} 1.5 & -0.5 \\ 0.25 & -0.5 \end{bmatrix} S$$

$Y_{11} = 1.5 \quad Y_{12} = -0.5$

$Y_{21} = 0.25 \quad Y_{22} = -0.5$

$$[Z] = \begin{bmatrix} Y_{22}/\Delta Y & -Y_{12}/\Delta Y \\ -Y_{21}/\Delta Y & Y_{11}/\Delta Y \end{bmatrix} \Omega$$

$$[Z] = \begin{bmatrix} 0.8 & -0.8 \\ 0.4 & -2.4 \end{bmatrix} \Omega$$

$\Delta y = 1.5(-0.5) - (-0.5)(0.25)$

$\Delta y = -0.625$